

Project Title :

AI-Based RF Interference Detection Assistant using IBM watsonx Orchestrate

1. Domain

Artificial Intelligence + Telecommunication + Wireless Network Optimization

This project focuses on using AI to analyze wireless network signals, detect RF (Radio Frequency) interference, identify its possible source, and provide solutions to improve communication quality.

2. Problem Statement

Wireless communication systems such as Wi-Fi, Bluetooth, IoT devices, satellites, and 4G/5G networks often suffer from **RF interference**, where unwanted radio signals disturb the desired communication signals. This causes problems like slow internet speed, video call interruptions, packet loss, weak connectivity, and unstable network performance.

Currently, identifying RF interference requires manual monitoring and expert knowledge, making the process time-consuming and difficult. Therefore, an AI-based assistant is needed to automatically analyze RF parameters, detect interference patterns, find possible causes, and recommend corrective actions for improving network reliability.

3. Project Objective

The main objective of this project is to develop an AI-powered assistant that can:

- Monitor and analyze wireless signal parameters automatically.
- Detect abnormal RF interference conditions.
- Identify possible sources and causes of interference.
- Evaluate overall network health.
- Provide intelligent recommendations to reduce interference.
- Generate easy-to-understand RF health reports using AI agents.

4. Key Parameters Analyzed

The AI agent analyzes important RF/network parameters:

- **RSSI (Received Signal Strength Indicator):** Measures the strength of the received signal.
- **dBm :** Represents signal power level.
- **Noise Floor:** Measures unwanted background radio noise.
- **SNR (Signal-to-Noise Ratio):** Compares signal strength with noise level.
- **Frequency:** Identifies the operating frequency band.
- **Wi-Fi Channel:** Helps detect channel congestion and interference.

Example:

- Strong signal + high noise → Possible interference issue.
- Low SNR → Poor communication quality.

5. Working of the AI System

Step 1: Data Collection

The system receives RF/network data from devices or monitoring tools.

Input example:

- Frequency: 2.4 GHz
- Signal Strength: -86 dBm
- Noise Level: -81 dBm
- Channel: 6
- SNR: 5 dB

Step 2: Signal Monitoring Agent

This agent checks the received RF parameters and monitors network conditions.

Task:

- Collect signal information.
- Identify abnormal values.

Step 3: Interference Detection Agent

This AI agent analyzes the data to determine whether interference exists.

Task:

- Compare signal and noise levels.
- Detect poor SNR conditions.
- Identify interference patterns.

Step 4: Root Cause Analysis Agent

This agent determines the possible reason behind interference.

Possible causes:

- Channel congestion
- Nearby electronic devices
- Signal overlap
- Weak signal strength
- Environmental noise

Step 5: Recommendation Agent

Provides solutions based on detected problems.

Examples:

- Change Wi-Fi channel.
- Reduce distance between device and router.
- Remove interference sources.
- Optimize network settings.

Step 6: Report Generation Agent

Creates a final RF Health Report containing:

- Network status
- Interference level
- Possible cause
- Recommended action
- Network improvement score

6. IBM watsonx Orchestrate Role

IBM watsonx Orchestrate acts as the coordinator of multiple AI agents.

It manages the complete workflow:

**RF Data Input → Signal Analysis → Interference Detection → Cause Identification
→ Solution Recommendation → Report Generation**

It helps different AI agents work together efficiently.

7. Challenges

- Accurate detection of interference in changing environments.
- Differentiating normal signal variation from actual interference.
- Identifying the exact source of interference.
- Providing real-time analysis.
- Integrating multiple AI agents effectively.
- Giving simple explanations for technical RF problems.

8. Expected Output

The system provides:

RF Health Report

Example:

Status: Poor Network Condition

Detected Issue: High RF Noise

Cause: Possible channel interference

SNR: 5 dB (Low)

Recommendation: Switch to less crowded channel and reduce interference sources.

9. Applications

- Home Wi-Fi optimization
- Enterprise network monitoring
- IoT communication systems
- 4G/5G network maintenance
- Satellite communication
- Smart city wireless networks

This is the complete project overview for your **AI-Based RF Interference Detection Assistant using IBM watsonx Orchestrate** .